Programming Fundamentals – Lab 04 Tasks

<u>Important:</u> Always check that your inputs are valid before doing any calculations. Use the right formulas, choose the proper data types, and apply decision structures (like if/else or switch) to handle different cases correctly. This ensures accurate results and avoids logical mistakes. Do proper calculations in the **Tasks 05, 06, 07 and 08** with proper formulas.

Task 1: Sports Team Selection Eligibility

Scenario

A college selects students for the annual sports tournament only if their practice attendance is 75% or above.

Requirements

- **Input:** practice attendance percentage
- Attendance ≥ 75 → Selected for Tournament
- Else → Not Selected

Concept Used

if – else

code:

```
int main(){
    int attendance;
    printf("enter your attendance percentage ? =");
    scanf("%d", &attendance);
    if(attendance>=75){
        printf("selected for tournament ");
    }
        else{
            printf("not selected ");
        }

    return 0;}.
```

Task 2: Internet Data Usage Category

Scenario

An internet service provider categorizes users based on monthly data consumption.

Requirements

- **Input:** total GB used
- Conditions:
  - Data ≤ 50 GB → Basic User
  - Data 51–150 GB → Standard User
  - Data > 150 GB → Heavy User

Concept Used

if – else if – else

code:

```
int main(){
    float internet;
    printf("enter total gb used ? =");
    scanf("%f", &internet);
    if(internet<=50){
        printf("basic user");
    }
    else if (internet>=51 && internet<=150){
        printf("standard user");
    }
        else{
            printf("heavy user ");
        }

    return 0; }
```

Task 3: Bank Account Balance Status

Scenario

A banking system determines the status of a user’s account balance.

Requirements

- **Input:** account balance (integer)
- Balance > 0 → Positive Balance
- Balance < 0 → Overdrawn
- Balance = 0 → Zero Balance

Concept Used

if – else if – else

code:

```
int main(){
        int balance;
        printf("enter account balance ? =");
        scanf("%d", &balance);
        if(balance>0){
                printf("balance is positive");
        }
        else if (balance<0){
                printf("balance is overdrawn");
        }
                else{
                        printf("zero balance");
                }

        return 0;
```

Task 4: Wi-Fi Access Authentication System

Scenario

A secure Wi-Fi network grants access only if correct login credentials are entered.

Requirements

- **Predefined credentials:**
  - Username: user
  - Password: 7890
- **Input:** username and password
- If both match → Connected Successfully
- Else → Connection Failed

Concept Used

Logical operators (&&) with if – else

Code:

```
int main(){
	char u='u';
	char username;
	int i=6;
	printf("enter user name = ");
	scanf("%c", &username);
	printf("enter password= ");
	scanf("%d", &i);

	if(u==username && i==6547){
		printf("connected successfully");
	}
	else{
		printf("connection failed ");
	}
	return 0;
}
```

Task 5: Banking Service Menu System

Scenario

A bank provides different services to customers through an ATM system.

Menu

1. Balance Inquiry
2. Withdraw Money
3. Deposit Money
4. Exit

Requirements

- Input user choice
- Use **switch statement** to display relevant message
- Handle invalid choice using **default**

Concept Used

switch statement

code:

```
int menu;
printf("MENU:\n ");
printf("1. balance inquiry\n");
printf("2. withdraw money\n");
printf("3. deposit money\n ");
printf("4. exit\n");
printf("select a option: ");
scanf("%d", &menu);
switch(menu){
    case 1:
        printf("you selected balance inquiry");
        break;
```

```
        case 2:
                printf("you selected withdraw money");
                break;
        case 3:
            printf("you selected deposit money");
            break;
        case 4:
                printf("you selected exit");
                break;
        default :
                printf("invalid");
```

Task 6: Scholarship Evaluation System

Scenario

A university awards scholarships based on marks obtained in five subjects.

Requirements

- **Input:** Marks (0–100) for any 5 subjects
- Calculate percentage
- Percentage ≥ 85 → Full Scholarship
- Percentage ≥ 70 → Partial Scholarship
- Percentage ≥ 50 → Eligible for Consideration
- Percentage < 50 → Not Eligible

Concept Used

if – else if – else

code:

```
int main() {
   int m1, m2, m3, m4, m5;
   float total, percentage;
```

```c
printf("Enter marks of 5 subjects (0-100):\n");
scanf("%d %d %d %d %d", &m1, &m2, &m3, &m4, &m5);

total = m1 + m2 + m3 + m4 + m5;
percentage = (total / 500) * 100;

printf("Percentage = %.2f%%\n", percentage);

if (percentage >= 85) {
    printf("Full Scholarship\n");
}
else if (percentage >= 70) {
    printf("Partial Scholarship\n");
}
else if (percentage >= 50) {
    printf("Eligible for Consideration\n");
}
else {
    printf("Not Eligible\n");
}
```

Task 7: Festival Sale Discount Calculator

Scenario

A shopping mall offers discounts during a festival sale based on total purchase amount.

Requirements

- **Input:** total purchase amount
- ≥ 5000 → 20% discount

- ≥ 3000 → 10% discount
- < 3000 → No discount

Concept Used

if – else if – else

code:

```
int main() {
    float amount, discount = 0, finalAmount;

    printf("Enter total purchase amount: ");
    scanf("%f", &amount);

    if (amount >= 5000) {
        discount = amount * 0.20;
    }
    else if (amount >= 3000) {
        discount = amount * 0.10;
    }
    else {
        discount = 0;
    }

    finalAmount = amount - discount;

    printf("Discount = %.2f\n", discount);
    printf("Final Amount = %.2f\n", finalAmount);
```

Task 8: Math Operations Console Using Switch

Scenario

A math operations console performs different calculations based on user selection.

Menu

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Square of a number
6. Cube of a number
7. Square Root of a number
8. Power (x^y)
9. Absolute Value
10. Modulus (Remainder of two numbers)

Requirements

- Display menu
- Input user choice
- Use **switch statement**

Input Rules

- Options **1–4, 8, and 10** → input two numbers
- Options **5–7 and 9** → input one number

Validations

- Handle division by zero
- Handle square root of negative number
- Handle modulus by zero
- Handle invalid menu choice using **default**

Concept Used

- switch
- if (validation)
- Relational and logical operators

```
#include <stdio.h>
# include <math.h>
int main() {

    int choice;
    double num1, num2;

    printf("1. Addition\n");
    printf("2. Subtraction\n");
    printf("3. Multiplication\n");
    printf("4. Division\n");
    printf("5. Square of a number\n");
    printf("6. Cube of a number\n");
    printf("7. Square Root of a number\n");
    printf("8. Power (x^y)\n");
    printf("9. Absolute Value\n");
    printf("10. Modulus (Remainder of two numbers)\n");

    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch(choice) {

        case 1:
        case 2:
        case 3:
```

```
case 4:
case 8:
    printf("Enter two numbers: ");
    scanf("%lf %lf", &num1, &num2);

    if(choice == 1)
        printf("Result = %.2lf\n", num1 + num2);

    else if(choice == 2)
        printf("Result = %.2lf\n", num1 - num2);

    else if(choice == 3)
        printf("Result = %.2lf\n", num1 * num2);

    else if(choice == 4) {
        if(num2 != 0)
            printf("Result = %.2lf\n", num1 / num2);
        else
            printf("Error! Division by zero.\n");
    }

    else if(choice == 8)
        printf("Result = %.2lf\n", pow(num1, num2));

    break;

case 5:
    printf("Enter a number: ");
    scanf("%lf", &num1);
```

```
    printf("Square = %.2lf\n", num1 * num1);
    break;

case 6:
    printf("Enter a number: ");
    scanf("%lf", &num1);
    printf("Cube = %.2lf\n", num1 * num1 * num1);
    break;

case 7:
    printf("Enter a number: ");
    scanf("%lf", &num1);

    if(num1 >= 0)
        printf("Square Root = %.2lf\n", sqrt(num1));
    else
        printf("Error! Square root of negative number.\n");

    break;



case 9:
    printf("Enter a number: ");
    scanf("%lf", &num1);
    printf("Absolute Value = %.2lf\n", fabs(num1));
    break;

case 10: {
    int a, b;
```

```c
        printf("Enter two integers: ");
        scanf("%d %d", &a, &b);

        if(b != 0)
            printf("Remainder = %d\n", a % b);
        else
            printf("Error! Modulus by zero.\n");

        break;
    }

    default:
        printf("Invalid choice! Please select between 1 and 10.\n");
}

return 0;
}
```